SWE 101 : Cross Platform Development

Computing Technologies Department

<Practical Report <2>>

React Native Development Environment and Create a Basic Multi-Screen Application

<Submitted by>

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1. Aim

To design and implement responsive layouts in a React Native application using Expo and integrate navigation so that the user interface adapts to different screen sizes and orientations.

2. Objective

The goals of this exercise are:

- To apply Flexbox for responsive UI design.
- To use percentage-based dimensions and flexible layouts.
- To implement scrollable views for small screens.
- To use useWindowDimensions for runtime responsiveness.
- To integrate navigation between multiple screens.

3. Learning Outcome

After completing this practical, the learner is able to:

- Create responsive mobile applications using React Native and Expo.
- Design adaptable layouts using Flexbox properties.
- Implement navigation between multiple screens.
- Use ScrollView to handle overflow content.
- Write modular and clean React Native code.

4. Introduction

Today's mobile apps need to work properly on phones and tablets of different sizes. Because of this, we use responsive design to make sure the layout adjusts automatically and still looks good on every screen.

In React Native, we can achieve this using Flexbox, flexible layouts, ScrollView, and useWindowDimensions to change the UI based on screen size. In this practical, we build a simple Expo app with navigation between screens and responsive layouts to make the app user-friendly on all devices.

5. Requirements

Software Requirements:

- Node.js (latest version)
- npm
- Expo Go app
- Visual Studio Code

Hardware Requirements:

- Laptop with internet access
- Smartphone

6. Procedure

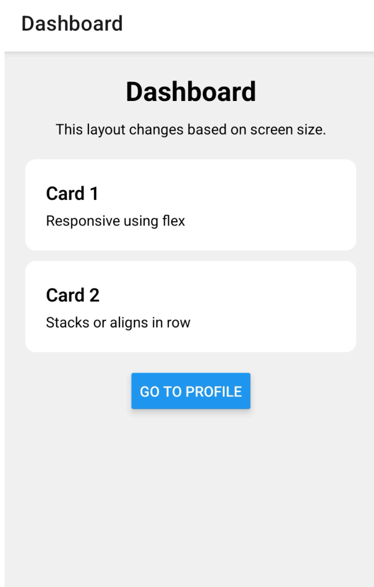
1. Verified Node.js and npm installation using terminal commands.
2. Created a new Expo project using:
   npx create-expo-app Practical2App
3. Installed React Navigation dependencies.
4. Created two screens:
   - Dashboard Screen
   - Profile Screen
5. Implemented navigation using Stack Navigator.
6. Designed responsive layouts using:
   - Flexbox (flex, flexDirection)
   - ScrollView
   - useWindowDimensions
7. Applied breakpoint logic to adjust layout for larger screens.
8. Tested the application on different screen sizes and orientations.
9. Modified layout to prevent overlapping and improve usability.

7. Program / Code

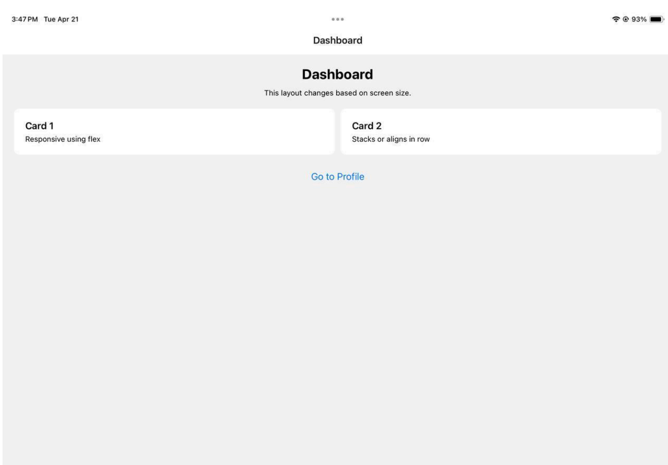
Repository Link**:** **https://github.com/Ldorji705/SWE_101_02240349.git**

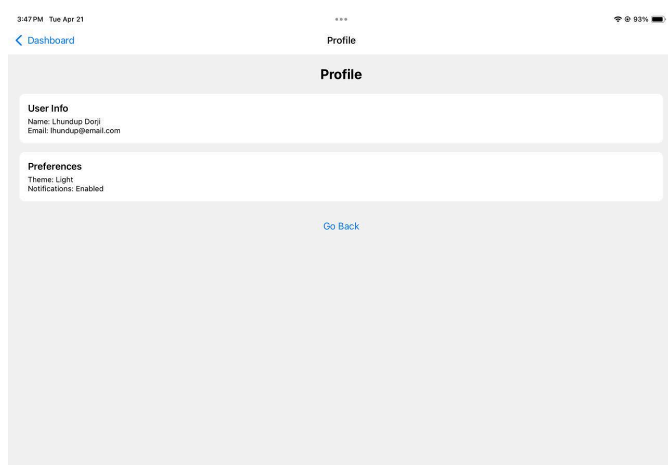
8.Output

Screenshot 1: Dashboard (Mobile View)

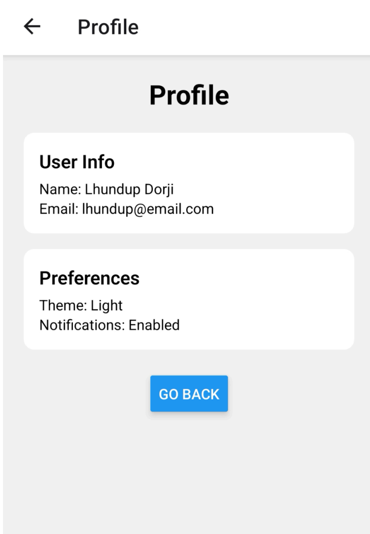


Screenshot 2: Dashboard (Tablet / Landscape View)





Screenshot 3: Profile Screen



9. Observation

During this practical, I observed that:

- The application runs successfully using Expo.
- Navigation between screens works correctly.
- Layout adapts based on screen width using useWindowDimensions.
- Cards are stacked vertically on small screens and aligned horizontally on larger screens.
- ScrollView prevents content from being cut off on smaller devices.
- No overlapping or layout breaking issues observed after adjustments.

10. Problem Encountered

I had a few small problems while doing the practical:

- Initial difficulty in setting up React Navigation.
- The layout did not adjust properly before using Flexbox correctly.
- Content overflow issues before implementing ScrollView.
- Breakpoint logic needed adjustment for better responsiveness.

11. Conclusion

The practical was successfully completed. A responsive React Native application was developed using Expo, demonstrating effective use of Flexbox, ScrollView, and useWindowDimensions. Navigation between screens was implemented smoothly, and the UI adapts well to different screen sizes and orientations.